Proximal Methods in Numerical Optimization

Lecture I – Motivation and Background

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⚠ these slides are under development: please email me for corrections and suggestions ⚠

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Background & Education

- Visiting Research Associate, Curtin University, Perth, AU 2022+2025
- Dr. rer. nat., Applied Mathematics UniBw M, Munich, DE 2021
- M.Sc., Mechatronics Engineering Univ. of Trento, IT 2016
- B.Sc., Industrial Engineering Univ. of Trento, IT 2014

Research Interests

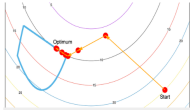
- Nonsmooth & nonconvex optimization
- Constrained structured optimization
- Optimal control & model predictive control
- Numerical analysis, scientific computing

PhD thesis: Augmented Lagrangian and Proximal Methods for Constrained Structured Optimization

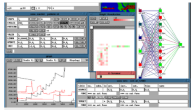
Engineering Mathematics Group at UniBw Munich

Prof. Matthias Gerdts, Institute within the **Department of Aerospace Engineering**

- **Optimal control** of ODEs, DAEs & PDEs
- **Model predictive control** real-time & online optimization
- **Nonsmooth & constrained optimization** proximal methods, augmented Lagrangians
- **Sensitivity analysis** & parameter identification
- **Trajectory optimization** aerospace, UAVs, autonomous vehicles
- **Multi-agent & cooperative control** generalized Nash equilibria, scheduling
- **Machine learning** for control imitation NMPC, neural networks



Optimization and Control



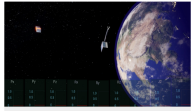
Problem-informed Machine Learning



Automated Driving



Flight Path Optimization



Space Robotics



Mobile Robots Lab

- Vehicle-in-the-Loop
- Flight Lab / Fixed-wing UAVs
- Multi-domain unmanned vehicles
- Industrial process control

What is this seminar all about?

Adapted from Parikh & Boyd's **Proximal Algorithms** [PB14]:

> *This [seminar] is about a class of* **optimization** *algorithms called* **proximal algorithms**. *Much like Newton's method is a standard tool for solving unconstrained smooth optimization problems of modest size, proximal algorithms can be viewed as an analogous* **tool for nonsmooth, constrained, large-scale, or distributed** *versions of these problems. They are very generally applicable, but are especially well-suited to problems of substantial recent interest involving large or high-dimensional datasets.* **Proximal methods sit at a higher level of abstraction than classical algorithms** *like Newton's method:* **the base operation is evaluating the proximal operator** *of a function, which itself involves solving a small [possibly non]convex optimization problem. [...]*

Part I Motivation and Background

Part II Proximal Point Methods

Part III Proximal Gradient Methods

Part IV Nonsmooth Problems with Constraints

Material online at
`aldma.github.io`

Outline

Motivating Example

Regularization

Applications

First-Order Optimization Methods

Proximal Operator

Least Square Regression

Setup

Let $m \in \mathbb{N}_{++}$. For $i = 1, \ldots, m$, given $x_i \in \mathbb{R}$:

$$y_i = ax_i + b + \epsilon_i$$

with ϵ_i random noise.

Matrix-vector form:

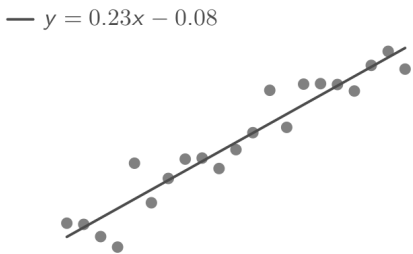
$$A = \begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_m & 1 \end{bmatrix}, \quad x = \begin{bmatrix} a \\ b \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}, \quad \epsilon = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_m \end{bmatrix}.$$

$$\implies \quad y = Ax + \epsilon.$$

Least Square Regression – Solution

Least square regression: estimating x from y

$$\text{Model } y = Ax + \epsilon \quad \Longrightarrow \quad \underset{x}{\text{minimize}} \quad \|Ax - y\|^2$$



Full column rank case:

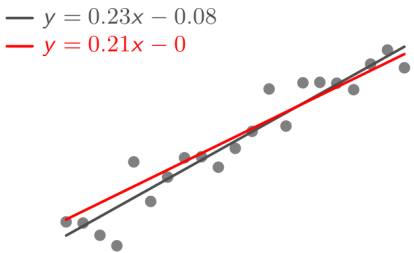
$\text{minimize}_x \|Ax - y\|^2 \iff A^\top Ax = A^\top y \iff x = (A^\top A)^{-1} A^\top y.$

$A^\top A$ **not invertible** $\Longrightarrow$ gradient descent: $x^{(k+1)} = x^{(k)} - \gamma_k A^T (Ax^{(k)} - y) \to x^\star.$

Non-negative Least Square

Non-negative least square regression

$$\underset{x}{\text{minimize}} \quad \|Ax - y\|^2 \quad \text{such that} \quad x \geq 0.$$



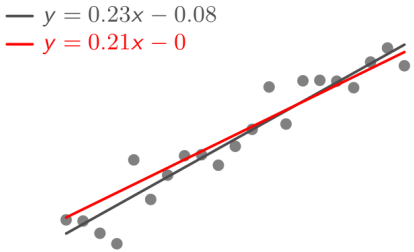
Smooth problem becomes **non-smooth** due to $x \geq 0$.

Jingwei Liang's "Selected Topics in Non-smooth Optimization"

Sparse Least Square

Sparse least square regression (constraint)

$$\underset{x}{\text{minimize}} \quad \|Ax - y\|^2 \quad \text{such that} \quad x \text{ has few nonzero components.}$$



Sparse least square regression (penalty)

$$\underset{x}{\text{minimize}} \quad \|Ax - y\|^2 + \text{penalty}(\text{nonzeros of } x).$$

Regularization

Regularized Least Square

Let $R(x)$ be a function promoting prior information, e.g. non-negativity or other simple constraints.

Regularized least square

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad \frac{1}{2} \|f(x)\|^2 + \mu R(x)$$

- $\mu > 0$ provides a balance between *simplicity* and *fidelity*.
- The choice of $R(x)$ depends on the prior information.

Regularization

$R(x)$ forces the solution to be "simpler", "more robust", "more expressive":

- Obtain results for ill-posed problems (*e.g.* image processing).
- Prevent overfitting (*e.g.* machine learning).

Problems min $f + g$ are often called "composite" in the literature...

A More General Formulation

General regularized problem

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad F(x) + R(x) \quad \text{subject to} \quad \ell \leq c(x) \leq u$$

Choices of F: quadratic loss, logistic loss, squared hinge loss, ...

Choices of R: ℓ_1-norm, ℓ_2-norm, ℓ_∞-norm, MCP, SCAD, rank, nuclear norm, ...

- Without R, back to nonlinear programming.
- Without explicit constraints, back to structured problems.

Penalties and Constraints

"It's convenient for many purposes to consider functions f that are allowed to be **extended-real-valued**, i.e., to take values in $\overline{\mathbb{R}} := [-\infty, \infty]$ instead of just $\mathbb{R}$"

Example: **indicator** of a set $C \subset \mathbb{R}^n$:

$$\delta_C : \mathbb{R}^n \to \overline{\mathbb{R}}, \quad \delta_C(x) := \begin{cases} 0 & \text{if } x \in C, \\ \infty & \text{otherwise.} \end{cases}$$

From minimizing over C

$$\underset{x \in C}{\text{minimize}} \quad f(x) \qquad \text{(explicit constraint)}$$

to minimizing over all of $\mathbb{R}^n$

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad \hat{f}(x) := f(x) + \delta_C(x) \qquad \text{(implicit constraint)}$$

"This helps in thinking abstractly about minimization and in achieving a single framework for the development of properties and results"

Rockafellar & Wets "Variational Analysis" [RW98, §1.A]

Applications

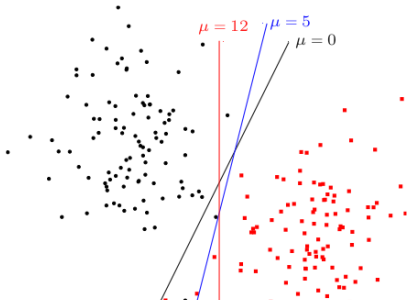
Example: Machine Learning – Sparse Logistic Regression

Sparse logistic regression

Let $(A_i, b_i) \in \mathbb{R}^n \times \{\pm 1\}$, $i = 1, \ldots, m$:

$$\underset{(x,y)\in\mathbb{R}^n\times\mathbb{R}}{\text{minimize}} \quad \frac{1}{m}\sum_{i=1}^{m} f(x^T A_i + y;\ b_i) + \mu \|x\|_1$$

where $f(u_i;\ b_i) = \log(1 + e^{-u_i b_i})$.



Effect of μ:

$\mu = 0$: original logistic fit
$\mu = 5$: moderate sparsity
$\mu = 12$: stronger regularization

sparser $\implies$ less memory, faster inference

Examples: Total Variation

Total Variation (TV) [ROF92]

Let ∇ be the discrete gradient operator:

$$\mathrm{TV}(x) := \|\nabla x\|_1$$



Original image



Horizontal gradient



Vertical gradient

Medical Imaging – MRI Reconstruction

Observation model:

$$f = \mathcal{F}x + \varepsilon$$

where x is piecewise constant/smooth and $\mathcal{F}$ is partial Fourier transform.

TV-based MRI reconstruction

Let $p \in \{1, 2\}$:

$$\underset{x \in \mathbb{R}^{m \times n}}{\text{minimize}} \quad \frac{1}{2} \|\mathcal{F}x - f\|_p^p + \mu \|\nabla x\|_1$$

Video Decomposition

Observation model:

$$y_{\text{obs}} = \bar{B} + \bar{F} + \varepsilon$$

where $\bar{B}$ is the background (prior: low rank $\rightsquigarrow$ nuclear norm, rank constraint), $\bar{F}$ is the foreground (prior: sparse $\rightsquigarrow$ ℓ_1-norm).

Principal component pursuit [CLMW11]

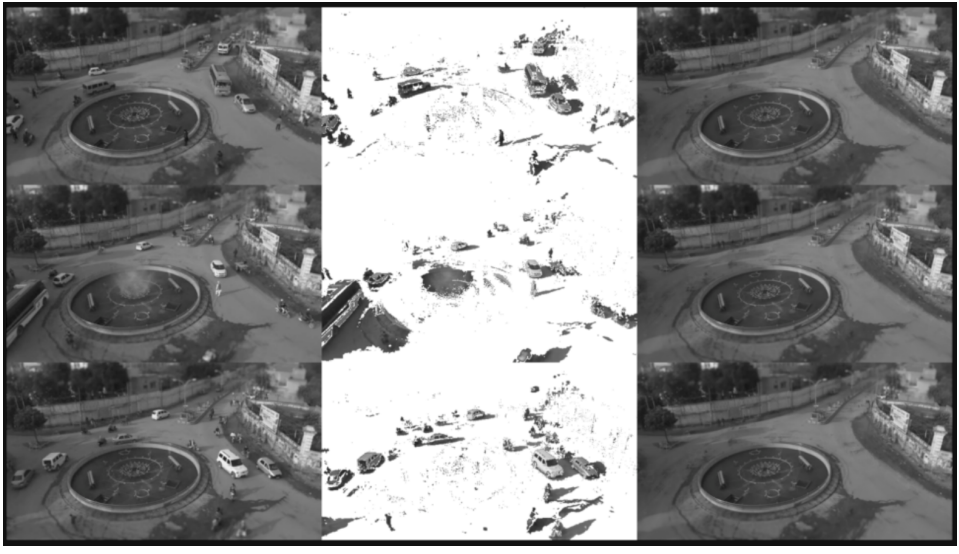
$$\underset{B,F \in \mathbb{R}^{m \times n}}{\text{minimize}} \quad \frac{1}{2} \left\| B + F - y_{\text{obs}} \right\|^2 + \mu \left\| B \right\|_* + \nu \left\| F \right\|_1$$

Rank-constrained decomposition [ASPvW20]

$$\underset{B,F \in \mathbb{R}^{m \times n}}{\text{minimize}} \quad \frac{1}{2} \left\| B + F - y_{\text{obs}} \right\|^2 + \nu \left\| F \right\|_1 \quad \text{subject to} \quad \text{rank}(B) \leq r.$$

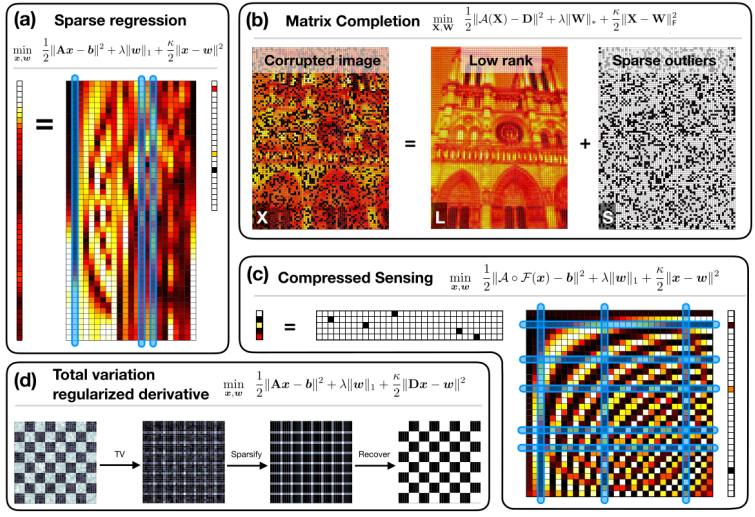
VideoBackgroundRemoval.ipynb in StructuredOptimization.jl

Video decomposition



VideoBackgroundRemoval.ipynb in StructuredOptimization.jl

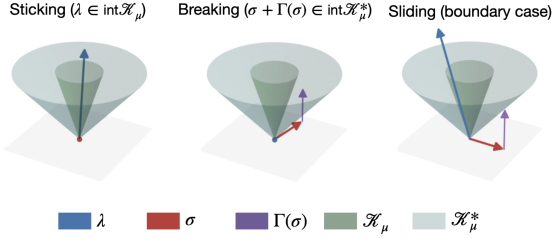
. . . and many more examples

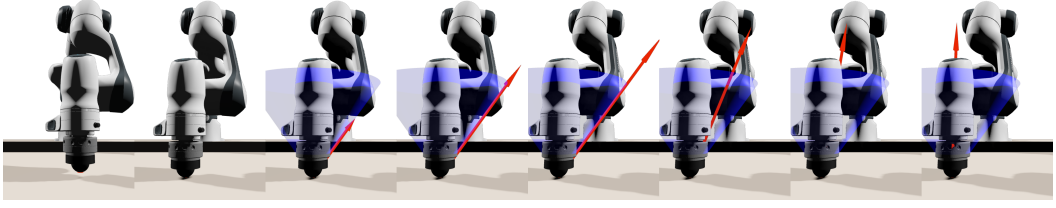


Zheng et al. "*A Unified Framework for Sparse Relaxed Regularized Regression: SR3*" [pysr3]

Robotics

Inverse dynamics and trajectory optimization with **frictional contact**





First-Order Optimization Methods

Non-smooth Optimization

Non-smooth optimization problem

Let $r \in \mathbb{N}_{++}$:

$$\underset{x_1,\ldots,x_r}{\text{minimize}} \left\{ \Phi(x) \coloneqq F(x_1, \ldots, x_r) + \sum_{i=1}^{r} R_i(K_i x_i) \right\}$$

where F is smooth (objective/loss/data fidelity),
R_i is prox-friendly (regularization),
and K_i are linear/nonlinear operators.

Non-smooth, (non-convex), composite, high dimension

First-Order Methods: Basic Ingredients

First-Order Methods: Numerical schemes that use *at most* first-order oracles of the objective (gradient, subgradient, prox, ...)

Gradient Descent [Cauchy 1847]

$$\underset{x\in\mathbb{R}^n}{\text{minimize}} \quad F(x)$$

F convex smooth, ∇F L-Lipschitz.

GD update:

$$x^{(k+1)} = x^{(k)} - \gamma_k \nabla F(x^{(k)}), \quad \gamma_k \in \left(0, \tfrac{2}{L}\right).$$

where ∇F is the **gradient** of F.

Proximal Point Algorithm [Roc76b]

$$\underset{x\in\mathbb{R}^n}{\text{minimize}} \quad R(x)$$

R proper, lsc, convex.

PPA update:

$$x^{(k+1)} = \text{prox}_{\gamma_k R}(x^{(k)}), \quad \gamma_k > 0.$$

where the **proximal operator** is

$$\text{prox}_R(v) := \arg\min_x \left\{ R(x) + \tfrac{1}{2}\|x - v\|^2 \right\}.$$

Problem Classes

Problem structure	**Method (convex case)**
$\min R_1 + R_2$	Douglas–Rachford splitting [DR56]
$\min_{x,z}\{R_1(x) + R_2(z) \mid Ax + Bz = c\}$	ADMM [GM75, GM76]
$\min F + R$	Forward–Backward splitting [LM79]
$\min F + R \circ K$	Primal–Dual splitting [CP11]
$\min F + \sum_i R_i$	Generalized FB splitting [RFP13]

More difficult to exploit structure in nonconvex settings, usually.

Settings and History: A New Hope

Douglas & Rachford [DR56], Moreau [Mor65], Rockafellar [Roc76a], Gabay & Mercier [GM76]

$$\underset{x}{\text{minimize}} \quad R(x), \qquad \underset{x}{\text{minimize}} \quad R_1(x) + R_2(x)$$

Lions & Mercier [LM79], Fukushima & Mine [FM81]

$$\underset{x}{\text{minimize}} \quad F(x) + R(x)$$

. . . forgotten for decades, until Beck & Teboulle's **accelerated prox-grad** [BT09]

Primal-dual: Chambolle & Pock [CP11]

$$\underset{x}{\text{minimize}} \quad F(x) + R(Kx)$$

Parikh & Boyd's FnT on "Proximal Algorithms" [PB14] **still convex**

Settings and History: Nonconvexity Awakens

Attouch, Bolte, Teboulle et al. [ABS13, BST14, BBT17, BSTV18]
Beck's book *First-Order Methods in Optimization* [Bec17]

$\rightsquigarrow$ Winter school on **"Modern methods in nonsmooth optimization"** in 2018 - Kanzow

Recent and fast-growing field [ABO22, DMT22, KM22, HT23, DMJKM23, DM23, DMT25, EPLP25, DFZ$^+$25, DHO26]

Relaxing assumptions, improving rates/complexity, weakening oracles, broader settings...

Convexity strikes back! Adaptive variants [MM24, LTSP25] for ML

Not in this course, but topics of active research

- Accelerated: same oracles, faster rates
- Decentralized / distributed
- Inexact evaluations
- Stochastic / randomized
- Riemannian / Bregman

Proximal Operator

Proximal Mapping – Definition

Definition – Proximal Mapping

The proximal mapping (proximity operator) of R with parameter $\gamma > 0$ is

$$\operatorname{prox}_{\gamma R}(y) := \operatorname*{arg\,min}_{x \in \mathbb{R}^n} \left\{ R(x) + \frac{1}{2\gamma} \|x - y\|^2 \right\}.$$

Function $R \colon \mathbb{R}^n \to \overline{\mathbb{R}} := \mathbb{R} \cup \{\infty\}$ should be proper and lsc.

Proximal methods adopt the proximal mapping as oracle to access R

- for **convex** R, $\operatorname{prox}_{\gamma R}(y)$ is nonempty and unique for all $y \in \mathbb{R}^n$ and $\gamma > 0$.
- for R **bounded from below**, $\operatorname{prox}_{\gamma R}(y)$ is nonempty for all $y \in \mathbb{R}^n$ and $\gamma > 0$.
- in general, the stepsize γ has to be small enough for $\operatorname{prox}_{\gamma R}$ to be well-defined.
- **prox-boundedness** of R: there exists $\gamma_R > 0$ such that $\operatorname{prox}_{\gamma R}(y)$ is nonempty for all $y \in \mathbb{R}^n$ and $\gamma \in (0, \gamma_R)$.

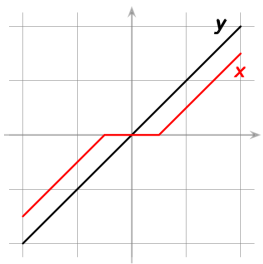
Example I: Prox operator

Absolute value

Let $R(x) := |x|$. Then

$$\operatorname{prox}_{\gamma|\cdot|}(x) = \operatorname*{arg\,min}_{y\in\mathbb{R}} \left\{ |y| + \frac{1}{2\gamma}(y-x)^2 \right\} = \begin{cases} 0 & \text{if } |x| \le \gamma, \\ x - \operatorname{sign}(x)\gamma & \text{if } |x| \ge \gamma. \end{cases}$$

Soft-thresholding operator:



Use symmetry and consider $y \ge 0$:

$$\min_{y\ge 0} \left\{ y + \frac{1}{2\gamma}(y-x)^2 \right\} \implies y = \max\{0, x-\gamma\}$$

Example II: Prox operator

Indicator of a convex set

Given $l, u \in \mathbb{R}$, $l \leq u$, let $R(x) \coloneqq \delta_{[\ell,u]}(x) \coloneqq \begin{cases} 0 & \text{if } x \in [\ell, u], \\ \infty & \text{otherwise} \end{cases}$. Then

$$\operatorname{prox}_{\gamma\, \delta_{[\ell,u]}}(x) = \operatorname*{arg\,min}_{y \in \mathbb{R}} \left\{ \delta_{[\ell,u]}(y) + \frac{1}{2\gamma}(y - x)^2 \right\} = \operatorname{proj}_{[\ell,u]}(x)$$

$$\operatorname{prox}_{\gamma\, \delta_{[\ell,u]}}(x) = \operatorname*{arg\,min}_{y \in \mathbb{R}} \left\{ \delta_{[\ell,u]}(y) + \frac{1}{2\gamma}(y - x)^2 \right\} = \operatorname*{arg\,min}_{y \in [\ell,u]} \frac{1}{2\gamma}(y - x)^2 = \begin{cases} \ell & \text{if } x \leq \ell, \\ y & \text{if } \ell \leq x \leq u, \\ u & \text{if } x \geq u. \end{cases}$$

Interpretations: Generalized Projection

Definition: Euclidean/Orthogonal projection

Given a closet nonempty set S,

$$\operatorname{proj}_S(y) := \operatorname*{arg\,min}_{x \in S} \|x - y\|.$$

Let δ_S be the indicator of S. Then

$$\begin{aligned}\operatorname{proj}_S(y) := \operatorname*{arg\,min}_{x \in S} \|x - y\| = \operatorname*{arg\,min}_{x \in S} \frac{1}{2}\|x - y\|^2 = \operatorname*{arg\,min}_{x \in S} \frac{1}{2\gamma}\|x - y\|^2 \\ = \operatorname*{arg\,min}_{x \in \mathbb{R}^n} \left\{ \delta_S(x) + \frac{1}{2\gamma}\|x - y\|^2 \right\} = \operatorname{prox}_{\gamma\,\delta_S}(y)\end{aligned}$$

for all $\gamma > 0$.

Interpretations: Smoothing

Definition – Moreau Envelope

Given $\gamma > 0$, the Moreau envelope (or Moreau–Yosida regularization) R^γ of R is

$$R^\gamma(y) := \inf_{x \in \mathbb{R}^n} \left\{ R(x) + \frac{1}{2\gamma} \|x - y\|^2 \right\}.$$

Also denoted $\mathrm{e}_\gamma R$ or $\mathrm{env}_\gamma R$ in the literature.

The Moreau envelope R^γ is a convolution of R with $\frac{1}{2\gamma}\|\cdot\|^2$: essentially a smoothed or regularized form of R.

- It has full domain $\mathbb{R}^n$, even when R does not.
- The sets of minima and minimizers of R and R^γ are the same.
- For **convex** R, it is continuously differentiable, even when R is not.

Proximal operator and Moreau envelope

Given R and $\gamma > 0$,

$$\operatorname{prox}_{\gamma R}(x) := \operatorname*{arg\,min}_{z \in \mathbb{R}^n} \left\{ R(z) + \frac{1}{2\gamma} \|z - x\|^2 \right\}, \qquad R^\gamma(x) := \inf_{z \in \mathbb{R}^n} \left\{ R(z) + \frac{1}{2\gamma} \|z - x\|^2 \right\}.$$

$\operatorname{prox}_{\gamma R}$ returns a point that actually achieves the infimum that defines R^γ

$$\forall x \in \mathbb{R}^n: \quad R^\gamma(x) = R(z) + \frac{1}{2\gamma} \|z - x\|^2 \quad \text{for all suff. small } \gamma > 0 \text{ and } z \in \operatorname{prox}_{\gamma R}(x).$$

For **convex** (or **prox-regular**) R, the gradient of the Moreau envelope is given by

$$\nabla R^\gamma(x) = \frac{x - \operatorname{prox}_{\gamma R}(x)}{\gamma}.$$

Example I: Moreau envelope

Absolute value

Let $R(x) := |x|$. Then

$$\operatorname{prox}_{\gamma|\cdot|}(x) = \begin{cases} 0 & \text{if } |x| \le \gamma, \\ x - \operatorname{sign}(x)\gamma & \text{otherwise.} \end{cases}$$

$$R^\gamma(x) = \inf_{y \in \mathbb{R}} \left\{ |y| + \frac{1}{2\gamma}(y-x)^2 \right\} = \begin{cases} \frac{x^2}{2\gamma} & \text{if } |x| \le \gamma, \\ |x| - \frac{\gamma}{2} & \text{otherwise.} \end{cases}$$

Since R is convex, R^γ is differentiable:

$$\nabla R^\gamma(x) = \begin{cases} \frac{x}{\gamma} & \text{if } |x| \le \gamma, \\ \operatorname{sign}(x) & \text{otherwise.} \end{cases}$$

[link to desmos]

Example II: Moreau envelope

Indicator of a convex set

Given $l, u \in \mathbb{R}$, $l \leq u$, let $R(x) := \delta_{[\ell,u]}(x) := \begin{cases} 0 & \text{if } x \in [\ell, u], \\ \infty & \text{otherwise} \end{cases}$. Then

$$\operatorname{prox}_{\gamma\, \delta_{[\ell,u]}}(x) = \operatorname{proj}_{[\ell,u]}(x)$$

$$R^\gamma(x) = \inf_{y \in [\ell,u]} \frac{1}{2\gamma}(y-x)^2 = \frac{1}{2\gamma} \begin{cases} (\ell - x)^2 & \text{if } x \leq \ell, \\ 0 & \text{if } \ell \leq x \leq u, \\ (u-x)^2 & \text{if } x \geq u, \end{cases} = \frac{1}{2\gamma} \operatorname{dist}^2_{[\ell,u]}(x).$$

Since R is convex, R^γ is differentiable:

$$\nabla R^\gamma(x) = \frac{x - \operatorname{proj}_{[\ell,u]}(x)}{\gamma}.$$

[link to desmos]

Interpretations: Gradient step on the Moreau envelope

For **convex** R, the gradient of the Moreau envelope is given by

$$\nabla R^\gamma(x) = \frac{x - \text{prox}_{\gamma R}(x)}{\gamma}.$$

Rewriting as

$$\text{prox}_{\gamma R}(x) = x - \gamma \nabla R^\gamma(x)$$

shows that $\text{prox}_{\gamma R}$ **can be viewed as a gradient step**, with stepsize γ, **for minimizing** R^γ (which has the same minimizers as R).

Calculus rules for prox

Quadratic perturbation

$$H(x) \coloneqq R(x) + \tfrac{\alpha}{2}\|x\|^2 + x^\top u + b,\ \alpha \geq 0 \implies \operatorname{prox}_H(y) = \operatorname{prox}_{R/(\alpha+1)}\left(\tfrac{y-u}{\alpha+1}\right).$$

Translation

$$H(x) \coloneqq R(x - z) \implies \operatorname{prox}_H(y) = z + \operatorname{prox}_R(y - z).$$

Scaling

$$H(x) \coloneqq R(x/\rho) \implies \operatorname{prox}_H(y) = \rho \operatorname{prox}_{R/\rho^2}(y/\rho).$$

Reflection

$$H(x) \coloneqq R(-x) \implies \operatorname{prox}_H(y) = -\operatorname{prox}_R(-y).$$

Composition ($K^{-1} = K^*$, bijective bounded linear)

$$H \coloneqq R \circ K \implies \operatorname{prox}_H(y) = K^* \operatorname{prox}_R(Ky).$$

Examples of prox

Projection $R(x) = \delta_S(x)$. Then $\text{prox}_{\gamma R}(x) = \text{proj}_S(x)$.

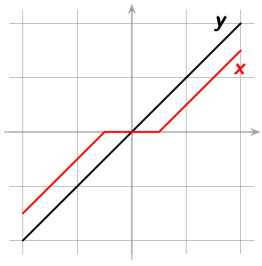
- Hyperplane $S = \{x : a^\top x = b\}$, $a \neq 0$: $\text{proj}_S(y) = y + \frac{b - a^\top y}{\|a\|^2} a$.
- Affine subspace $S = \{x : Ax = b\}$, $A \in \mathbb{R}^{m \times n}$, $\text{rank}(A) = m$: $\text{proj}_S(y) = y + A^\top (AA^\top)^{-1}(b - Ay)$.
- Nonnegative orthant $S = \mathbb{R}^n_+$: $\text{proj}_S(y) = \max\{0, y\}$ (elementwise).

Quadratic $R(x) = \frac{1}{2} x^\top Ax + b^\top x + c$, $A \succeq 0$:

$$\text{prox}_{\gamma R}(x) = (\mathbb{I} + \gamma A)^{-1}(x - \gamma b).$$

Examples of prox

Soft-thresholding operator:



Absolute value/ℓ_1 norm $R(x) = |x|$:

$$\operatorname{prox}_{\gamma R}(x) = \begin{cases} x - \gamma & \text{if } x \geq \gamma, \\ x + \gamma & \text{if } x \leq -\gamma, \\ 0 & \text{otherwise.} \end{cases}$$

Euclidean norm $R(x) = \|x\|_2$:

$$\operatorname{prox}_{\gamma R}(x) = \begin{cases} (1 - \frac{\gamma}{\|x\|})x & \|x\| > \gamma, \\ 0 & \text{otherwise.} \end{cases}$$

Nuclear norm $R(X) = \sum_i \sigma_i(X)$, $y = USV^\top$, $S = \operatorname{diag} \sigma$:

$$\operatorname{prox}_{\gamma R}(y) = U \operatorname{prox}_{\gamma \|\cdot\|_1}(S) V^\top.$$

Fancy some code?

Proximal **operators**

- proximity-operator.net (MATLAB/python)
- cvxpy.org (python)
- JuliaFirstOrder/ProximalOperators.jl (julia)

Proximal **algorithms**

- cvxpy.org (python)
- JuliaFirstOrder/ProximalAlgorithms.jl (julia)
- JuliaSmoothOptimizers/RegularizedOptimization.jl (julia)

Prox map encodes structure

Consider the LASSO problem:

$$\underset{x\in\mathbb{R}^n}{\text{minimize}} \quad \underbrace{\frac{1}{2}\|Ax-b\|^2}_{f(x)} + \underbrace{\lambda\|x\|_1}_{g(x)}$$

for some $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$ and $\lambda > 0$. It is well structured with f smooth and g prox-friendly.

The change of variable $x \equiv x_+ - x_-$ leads to the nonlinear program

$$\underset{x_+,x_-\in\mathbb{R}^n}{\text{minimize}} \quad f(x_+ - x_-) + \lambda 1^\top (x_+ + x_-) \quad \text{s.t.} \quad x_+, x_- \geq 0$$

which is ... larger and constrained.

De Simone et al. "*Sparse Approximations with Interior Point Methods*" (2022) [DSdSG+22]

Bareilles, Iutzeler & Malick "*Newton acceleration on manifolds identified by proximal gradient methods*" (2023) [BIM23]

Asking claude.ai (Sonnet 4.6)

Prompt:

Why are proximal methods important in nonsmooth optimization?

Reply:

The core difficulty they address. *Gradient descent breaks down when f is nonsmooth — the gradient may not exist, and even subgradient methods converge slowly and cannot be accelerated in general. Proximal methods sidestep this by replacing the gradient step with the* **proximal operator***:*

$$\text{prox}_{\alpha f}(x) = \arg\min_u \left\{ f(u) + \frac{1}{2\alpha}\|u - x\|^2 \right\}$$

This is always well-defined (strongly convex subproblem, unique minimizer) [☠] even when f has no gradient. Geometrically it finds the point that best trades off reducing f against staying near x.

They encode structure implicitly. *The power of the proximal framework is that for many nonsmooth functions that appear in practice, $\text{prox}_{\alpha f}(x)$ has* **closed-form solution***, so the "expensive" nonsmooth subproblem costs essentially nothing: [...]*

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